



CSE471 : System Analysis and Design
Lab Assignment on API Postman Collection

**Project Title : Clarity - B2B Invoice Discounting and Supply
Chain Finance Platform**

Group Number : 4

Backend Framework : Node js, Express js

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Section: 9

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APIs of Feature - 01

1. Get all invoice

feature 1 / all Invoices

GET http://localhost:5001/api/invoices

Send

Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

Key	Value	Description
Key	Value	Description

Body Cookies Headers (8) Test Results

200 OK • 1.12 s • 10.31 KB Save Response

JSON Preview Visualize

```
1 [
2   {
3     "id": "ef5c424b-792e-4488-8174-1c48835d81fb",
4     "invoice_number": "TESTDD-001",
5     "buyer_name": "Nova Retail Group",
6     "supplier_id": "1",
7     "status": "Payout Initiated",
8     "invoice_amount": "875000.00",
9     "payout_amount": "831250.00",
10    "file_url": null,
11    "frozen_at": null,
12    "due_date": "2026-09-03",
13    "submitted_date": "2026-08-20"
14  },
15  {
16    "id": "5a9f0aac-5a20-48b0-9275-586f76bf395b",
17    "invoice_number": "TESTDD-003",
18    "buyer_name": "Envoy Textiles",
19    "supplier_id": "1",
20    "status": "Buyer Confirmed",
21    "invoice_amount": "430000.00",
```

```
router.get('/', async (req, res) => {
  try {
    const result = await pool.query(
      `SELECT id, invoice_number, buyer_name, supplier_id, status,
        invoice_amount, payout_amount, file_url, frozen_at,
        TO_CHAR(due_date, 'YYYY-MM-DD') AS due_date,
        TO_CHAR(submitted_date, 'YYYY-MM-DD') AS submitted_date
      FROM invoices
      ORDER BY submitted_date DESC NULLS LAST, id DESC`
    );
    res.json(result.rows);
  } catch (error) {
    res.status(500).json({ message: 'Could not load invoices' });
  }
});
```

2. get specific invoice: <http://localhost:5001/api/invoices/:id>

feature 1 | get specific invoice

GET http://localhost:5001/api/invoices/ :id

Send

Docs Params Authorization Headers (6) Body Scripts Settings Cookies

Key	Value	Description	...	Bulk Edit
Key	Value	Description		

Path Variables

Key	Value	Description	...	Bulk Edit
id	ef5c424b-702e-4488-8174-1c48835d01fb	Description		

Body Cookies Headers (8) Test Results

200 OK • 850 ms • 567 B • Save Response

{ } JSON Preview Visualize

```

1 {
2   "id": "ef5c424b-702e-4488-8174-1c48835d01fb",
3   "invoice_number": "TEST00-001",
4   "buyer_name": "Nova Retail Group",
5   "supplier_id": "1",
6   "status": "Payout Initiated",
7   "invoice_amount": "875900.00",
8   "payout_amount": "831250.00",
9   "file_url": null,
10  "frozen_at": null,
11  "due_date": "2026-09-03",
12  "submitted_date": "2026-08-20"
13 }
```

```

router.get('/:id', async (req, res) => {
  try {
    const result = await pool.query(
      `SELECT id, invoice_number, buyer_name, supplier_id, status,
        invoice_amount, payout_amount, file_url, frozen_at,
        TO_CHAR(due_date, 'YYYY-MM-DD') AS due_date,
        TO_CHAR(submitted_date, 'YYYY-MM-DD') AS submitted_date
      FROM invoices WHERE id::TEXT = $1`,
      [req.params.id]
    );
    if (result.rowCount === 0) {
      return res.status(404).json({ message: 'No invoice with that id' });
    }
    res.json(result.rows[0]);
  } catch (error) {
    res.status(500).json({ message: 'Could not load that invoice' });
  }
});

```

3. Pipeline: http://localhost:5001/api/pipeline/invoices

feature 1 / pipeline

GET http://localhost:5001/api/pipeline/invoices Send

Docs Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

Key	Value	Description	...	Bulk Edit
Key	Value	Description		

Body Cookies Headers (8) Test Results 200 OK 1.10 s 8.35 KB Save Response

{} JSON Preview Visualize

```

1  [
2    {
3      "id": "ef5c424b-702e-4488-8174-1c48835d01fb",
4      "number": "TEST00-001",
5      "buyerName": "Nova Retail Group",
6      "amount": 875000,
7      "dueDate": "2026-09-03T00:00:00+00:00",
8      "currentStage": "Payout Initiated",
9      "history": [
10       {
11         "actor": "Supplier",
12         "stage": "Payout Initiated",
13         "timestamp": "2026-08-26T15:21:46.874754+00:00"
14       }
15     ]
16   },
17   {
18     "id": "25a9c38c-3666-4f4d-9b1a-797e68a0cc3a",
19     "number": "TEST00-002",

```

```

exports.getInvoiceStatus = async (req, res) => {
  const { id } = req.params;
  res.status(200).json({ message: `Fetch status for invoice ${id}`
});
};

```

4. get all notification: <http://localhost:5001/api/notifications>

feature 1 / get all notification

GET http://localhost:5001/api/notifications Send

Docs Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

Key	Value	Description	...	Bulk Edit
Key	Value	Description		

Body Cookies Headers (8) Test Results 200 OK 410 ms 6.91 KB Save Response

{ } JSON Preview Visualize

```

1  [
2    {
3      "id": 26,
4      "recipient": "buyer@clarityb2b.com",
5      "message": "[Watchlist] Supplier TEST-SUP slipped into the Distress band (score 40). Early warning on a supplier you are monitoring.",
6      "invoice_link": "/health",
7      "type": "distress_alert",
8      "is_read": true,
9      "created_at": "2026-08-15T03:03:28.858632+00:00"
10   },
11   {
12     "id": 25,
13     "recipient": "buyer@clarityb2b.com",
14     "message": "Supplier Supplier TEST-SUP has crossed into the Distress band (score 40).",
15     "invoice_link": "/health",
16     "type": "distress_alert",
17     "is_read": true,
18     "created_at": "2026-08-14T21:04:11.227396+00:00"
19   },
20   {
21     "id": 24,
22     "recipient": "buyer@clarityb2b.com",

```

```

exports.getNotifications = async (req, res) => {
  try {
    const recipient = req.query.recipient; // optional

    let query = supabase
      .from('notifications')
      .select('*')
      .order('created_at', { ascending: false });

    if (recipient) {
      query = query.eq('recipient', recipient);
    }

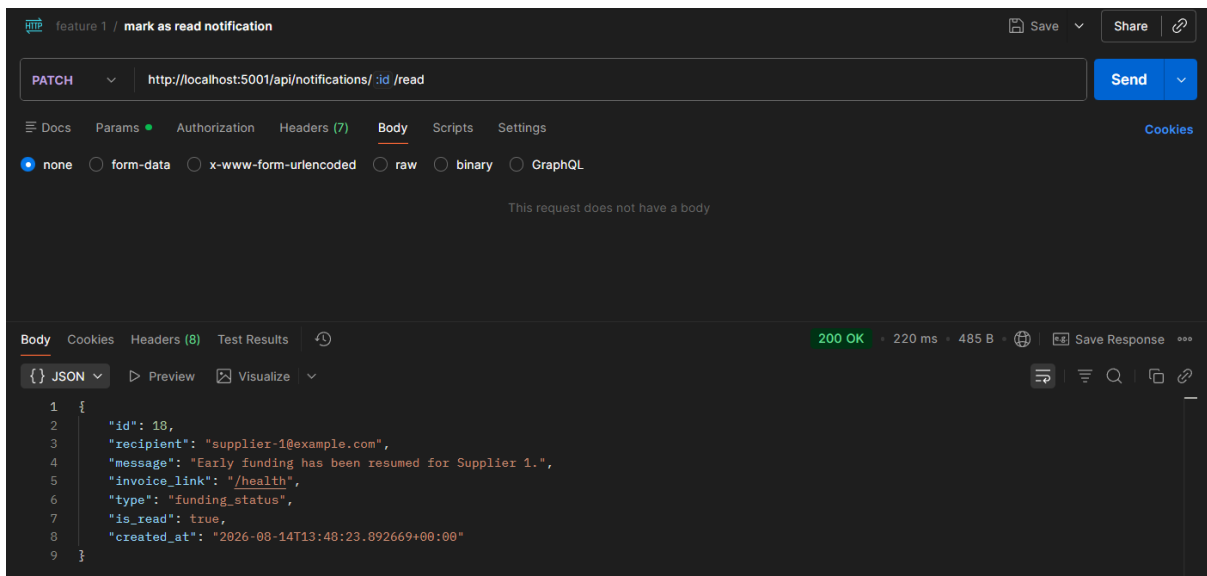
    const result = await query;

    if (result.error) {
      throw result.error;
    }

    res.status(200).json(result.data);
  } catch (error) {
    res.status(500).json({ error: error.message });
  }
};

```

5. mark as read notification: <http://localhost:5001/api/notifications/:id/read>



```
exports.markAsRead = async (req, res) => {
  try {
    const id = req.params.id;

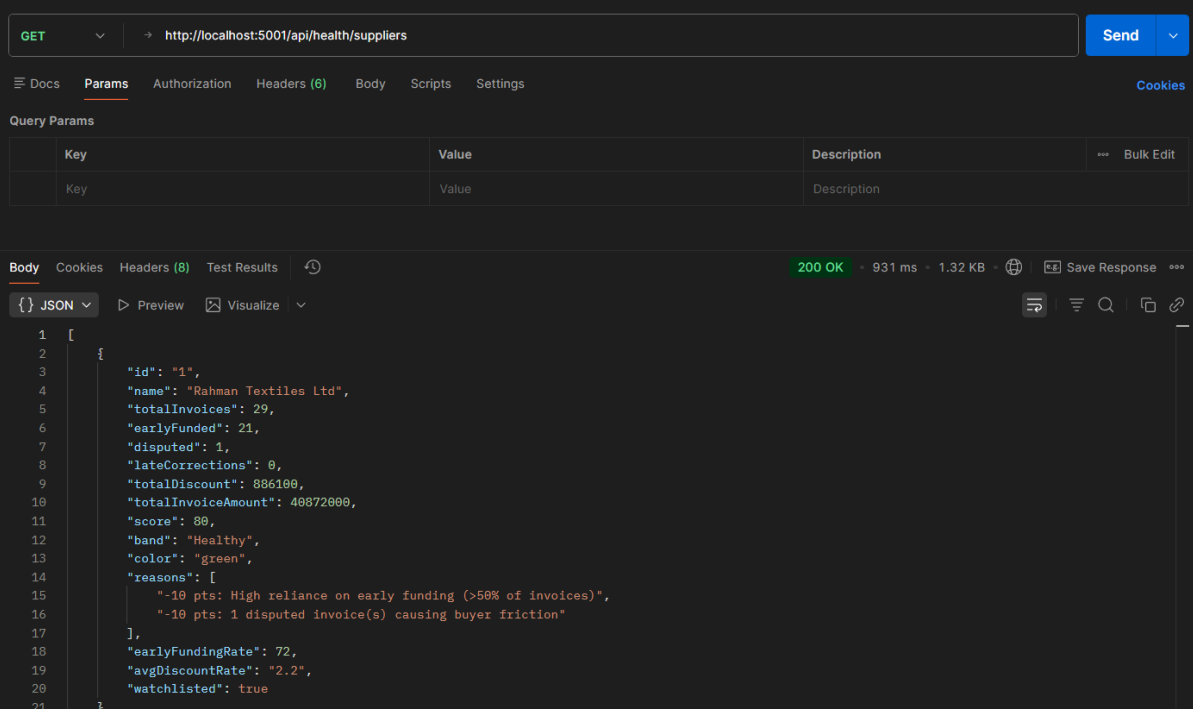
    const result = await supabase
      .from('notifications')
      .update({ is_read: true })
      .eq('id', id)
      .select()
      .single();

    if (result.error) {
      throw result.error;
    }

    res.status(200).json(result.data);
  } catch (error) {
    res.status(500).json({ error: error.message });
  }
};
```

APIs of Feature - 02

1. supplier health: <http://localhost:5001/api/health/suppliers>



The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** <http://localhost:5001/api/health/suppliers>
- Status:** 200 OK
- Response Time:** 931 ms
- Response Size:** 1.32 KB
- Body:** JSON

The JSON response body is as follows:

```
[
  {
    "id": "1",
    "name": "Rahman Textiles Ltd",
    "totalInvoices": 29,
    "earlyFunded": 21,
    "disputed": 1,
    "lateCorrections": 0,
    "totalDiscount": 886100,
    "totalInvoiceAmount": 49872000,
    "score": 80,
    "band": "Healthy",
    "color": "green",
    "reasons": [
      "-10 pts: High reliance on early funding (>50% of invoices)",
      "-10 pts: 1 disputed invoice(s) causing buyer friction"
    ],
    "earlyFundingRate": 72,
    "avgDiscountRate": "2.2",
    "watchlisted": true
  }
]
```

```
exports.getSupplierHealth = route(
  async (req, res) => {
    const suppliers = await
healthService.computeHealth(req.query.buyer);
    res.status(200).json(suppliers);
  },
  "Supplier Health Error",
  "Failed to compute supplier health",
);
```

2. supplier health recalculate: <http://localhost:5001/api/health/recalculate>

feature 1 / **supplier health recalculate** Save Share

POST http://localhost:5001/api/health/recalculate Send

Docs Params Authorization Headers (7) Body Scripts Settings Cookies

Query Params

Key	Value	Description	...	Bulk Edit
Key	Value	Description		

Body Cookies Headers (8) Test Results 200 OK 1.57 s 1.4 KB Save Response

JSON Preview Visualize

```

1 {
2   "message": "Supplier health recalculated and saved",
3   "suppliersProcessed": 3,
4   "data": [
5     {
6       "id": "1",
7       "name": "Rahman Textiles Ltd",
8       "totalInvoices": 29,
9       "earlyFunded": 21,
10      "disputed": 1,
11      "lateCorrections": 0,
12      "totalDiscount": 886100,
13      "totalInvoiceAmount": 40872000,
14      "score": 80,
15      "band": "Healthy",
16      "color": "green",
17      "reasons": [
18        "-10 pts: High reliance on early funding (>50% of invoices)",
19        "-10 pts: 1 disputed invoice(s) causing buyer friction"
20      ],
21      "earlyFundingRate": 72,

```

```

exports.recalculate = route(
  async (req, res) => {
    const suppliers = await healthService.runRecalculation();
    res.status(200).json({
      message: "Supplier health recalculated and saved",
      suppliersProcessed: suppliers.length,
      data: suppliers,
    });
  },
  "Recalculate Error",
  "Failed to recalculate supplier health",
);

```

3. **Summary:** <http://localhost:5001/api/health/summary>

GET http://localhost:5001/api/health/summary Send

Docs Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

Key	Value	Description	...	Bulk Edit
Key	Value	Description		

Body Cookies Headers (6) Test Results 200 OK 937 ms 381 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "totalSuppliers": 3,
3   "healthy": 3,
4   "watch": 0,
5   "distress": 0,
6   "healthyPercent": 100,
7   "watchPercent": 0,
8   "distressPercent": 0
9 }
```

```
exports.getSummary = route(
  async (req, res) => {
    const suppliers = await
healthService.computeHealth(req.query.buyer);
    res.status(200).json(healthService.summarize(suppliers));
  },
  "Supplier Summary Error",
  "Failed to compute supplier summary",
);
```

4. Watchlist for supplier: <http://localhost:5001/api/health/watchlist>

GET http://localhost:5001/api/health/watchlist Send

Docs Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers (8) Test Results 200 OK 863 ms 670 B Save Response

{} JSON Preview Visualize

```

1 [
2   {
3     "id": "1",
4     "name": "Rahman Textiles Ltd",
5     "totalInvoices": 29,
6     "earlyFunded": 21,
7     "disputed": 1,
8     "lateCorrections": 0,
9     "totalDiscount": 886100,
10    "totalInvoiceAmount": 40872000,
11    "score": 80,
12    "band": "Healthy",
13    "color": "green",
14    "reasons": [
15      "-10 pts: High reliance on early funding (>50% of invoices)",
16      "-10 pts: 1 disputed invoice(s) causing buyer friction"
17    ],
18    "earlyFundingRate": 72,
19    "avgDiscountRate": "2.2",

```

```

exports.getWatchlist = route(
  async (req, res) => {
    const suppliers = await healthService.computeHealth();
    res.status(200).json(suppliers.filter((supplier) =>
supplier.watchlisted));
  },
  "Get Watchlist Error",
  "Failed to fetch watchlist",
);

```